

Lecture 8

Saturday, February 28, 2026 9:17 AM

Today: Gradient Descent & Variants

What are we solving?

Supervised learning

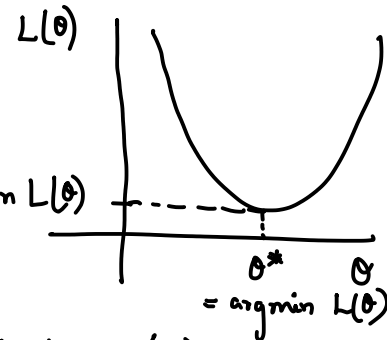
Loss function: measure of how wrong the model is.

$L(\theta)$

↑ ↖ parameters (β in your previous lecture)

Loss function (RSS)

$$\theta^* = \underset{\theta}{\operatorname{argmin}} L(\theta) \quad \min L(\theta)$$



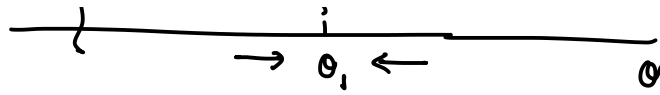
$\min L(\theta)$

$\operatorname{argmin} L(\theta)$: value of θ at which $L(\theta)$ achieves its minimum value

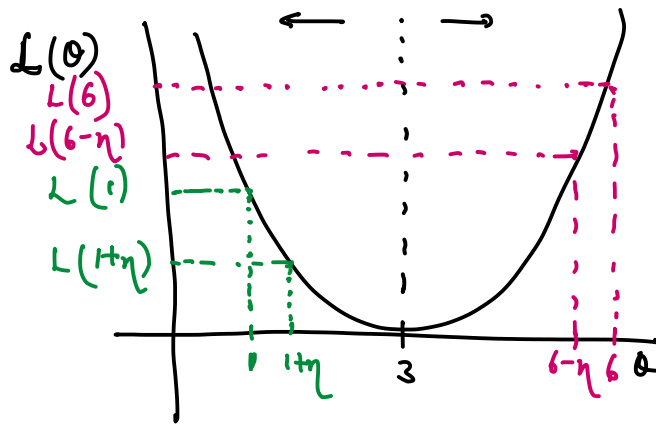
Core idea

- Closed-form solutions are often not possible
- Iterative algorithm to find θ^*
- Any differentiable loss function can be used in Gradient Descent





1D illustration



$$\begin{aligned} L(\theta) &= (\theta - 3)^2 \\ L'(\theta) &= 2(\theta - 3) \\ L'(0) &= 0 \\ \Rightarrow \theta &= 3 \\ L''(\theta) &= 2 \\ &\text{(basic calculus principles)} \end{aligned}$$

$$L(\theta) = (\theta - 3)^2 \quad \theta^* = \operatorname{argmin}_{\theta} L(\theta) = 3$$

$$\begin{aligned} \rightarrow L'(\theta) &= 2(\theta - 3) \\ L'(1) &= 2(-2) = -4 < 0 \\ \theta &= 1, \text{ move in the direction} \\ &\text{of the negative derivative} \\ \theta &= 1 + \eta \leftarrow \text{constant} \end{aligned}$$

$$\begin{aligned} L'(6) &= 2(3) = 6 \\ \theta &= 6, \text{ move in the direction of the negative derivative} \\ \theta &= 6 - \eta \leftarrow \text{constant} \end{aligned}$$

$$\begin{aligned} L'(3) &= 0 \\ \theta &= 3 \end{aligned}$$

What happens in higher dimensions?

$$\theta = (\theta_1 \ \theta_2 \ \dots \ \theta_d)^T \in \mathbb{R}^d$$

↑
set of real numbers

$$\begin{aligned} L : \mathbb{R}^d &\rightarrow \mathbb{R} \\ \downarrow &\quad \hookrightarrow \text{range / co-domain} \\ \text{domain} & \end{aligned}$$

Gradient : vector (d -dimensional)

$$\begin{aligned} L(\theta) : \mathbb{R}^d &\rightarrow \mathbb{R} \\ \nabla_{\theta} L(\theta) &= \begin{bmatrix} \frac{\partial L}{\partial \theta_1} \\ \vdots \\ \frac{\partial L}{\partial \theta_d} \end{bmatrix} \end{aligned}$$

$$\begin{pmatrix} \frac{\partial L}{\partial \theta_1} \\ \vdots \\ \frac{\partial L}{\partial \theta_d} \end{pmatrix} \in \mathbb{R}^d$$

$\frac{\partial L}{\partial \theta_i}$: partial derivative of L w.r.t. θ_i
 hold $\theta_1, \theta_2, \dots, \theta_{i-1}, \theta_{i+1}, \dots, \theta_d$ constant
 measure of how fast $L(\theta)$ changes when
 we change θ_i slightly holding all other
 dimensions constant

2-d examples

$$L(\theta) = \theta_1^2 + 4\theta_2^2 \quad (d=2)$$

$$\frac{\partial L(\theta)}{\partial \theta_1} = 2\theta_1 \quad \frac{\partial L(\theta)}{\partial \theta_2} = 8\theta_2$$

$$\nabla_{\theta} L(\theta) = \begin{pmatrix} 2\theta_1 \\ 8\theta_2 \end{pmatrix}$$

Fact (stated without proof)

$L(\theta)$: The direction of the steepest descent
 $\uparrow$
 $\mathbb{R}^d$ is $-\frac{\nabla L(\theta)}{\|\nabla L(\theta)\|}$

If you want to ~~at~~ take a small step of length ϵ ,
 then moving in the direction of $-\frac{\nabla L(\theta)}{\|\nabla L(\theta)\|}$

will give you the sharpest decrease.

$$\theta - 1D \quad -\frac{\nabla L(\theta)}{\|\nabla L(\theta)\|} = \begin{cases} -1 & \nabla L(\theta) > 0 \\ +1 & \nabla L(\theta) < 0 \end{cases} \quad \left. \begin{array}{l} \text{maps back} \\ \text{to 1D} \\ \text{solutions} \end{array} \right\}$$

Update rule for Gradient Descent

$$\theta \leftarrow \theta - \eta \nabla_{\theta} L(\theta)$$

$\uparrow$ update value of parameters
 $\uparrow$ current value of parameters
 $\uparrow$ step-size (> 0)
 $\nabla_{\theta} L(\theta)$ Gradient

$$L(\theta) = \theta_1^2 + 4\theta_2^2$$

$$\nabla L(\theta) = \begin{pmatrix} 2\theta_1 \\ 8\theta_2 \end{pmatrix}$$

$$\theta = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\nabla L(\theta) = \begin{pmatrix} 4 \\ 8 \end{pmatrix}$$

$$\theta \leftarrow \begin{pmatrix} 2 \\ 1 \end{pmatrix} - \eta \begin{pmatrix} 4 \\ 8 \end{pmatrix}$$

$$\theta = \begin{pmatrix} 2 - \eta 4 \\ 1 - \eta 8 \end{pmatrix}$$

Analogy : You are on a mountain

Effect of learning rate

- η too small $\rightarrow$ slow convergence
- η just right $\rightarrow$ smooth, quick convergence
- η large $\rightarrow$ oscillations, convergence
- η too large $\rightarrow$ divergence

Local vs global minima

- convex, $\rightarrow$ guaranteed to converge to global minima
- non-convex $\rightarrow$ no guarantees, but you will get to a local minima

Termination conditions

1. $\nabla_{\theta} L(\theta) = 0$: automatically stop
2. $\|\nabla L(\theta)\| < \epsilon = 10^{-6}$ (say)
 $\uparrow$
 fixed value

3. $|L(\theta_i) - L(\theta_{i-1})| < \delta = 10^{-6}$ (say) (loss not decreasing)
 $\uparrow$
 fixed value

$$\theta \leftarrow \theta - \eta \frac{1}{n} \sum (x\theta - y)$$

(work out the Gradient & verify this)

$X : n \times (d+1)$

large dataset $n = 10^6$